Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM: Generate Histogram of images, apply Histogram equalization and Histogram matching on it**	
Experiment No: 8	**Date:23-08-2026**	**Enrollment No:92400133121**

AIM: Generate Histogram of images, apply Histogram equalization and Histogram matching on it

Theory:

Histogram:

- A histogram is a graphical representation that shows the distribution of pixel intensities in an image.
- It displays the frequency of each intensity value on the horizontal axis and the number of pixels with that intensity value on the vertical axis.

Histogram Equalization:

- Histogram equalization is a technique used to enhance the contrast of an image by redistributing the intensity values.
- It stretches the histogram of an image to cover the full range of intensity values, resulting in a more balanced and visually appealing image.

Histogram Matching:

- Histogram matching is a process of transforming an image to match the histogram of a reference image.
- It is used to transfer the characteristics of the reference image's histogram to the target image, making them visually similar in terms of intensity distribution.

Flowchart:

- The flowchart for the program will consist of the following steps:
- Load the input image.
- Generate and display the histogram of the image.
- Apply histogram equalization to enhance the contrast.
- Generate and display the histogram of the equalized image.
- Load the reference image for histogram matching.
- Perform histogram matching between the input image and reference image.
- Generate and display the histogram of the matched image.
- Display the original image, equalized image, and matched image side by side.
- Save the equalized and matched images (optional).

CODE:

pip install scikit-image

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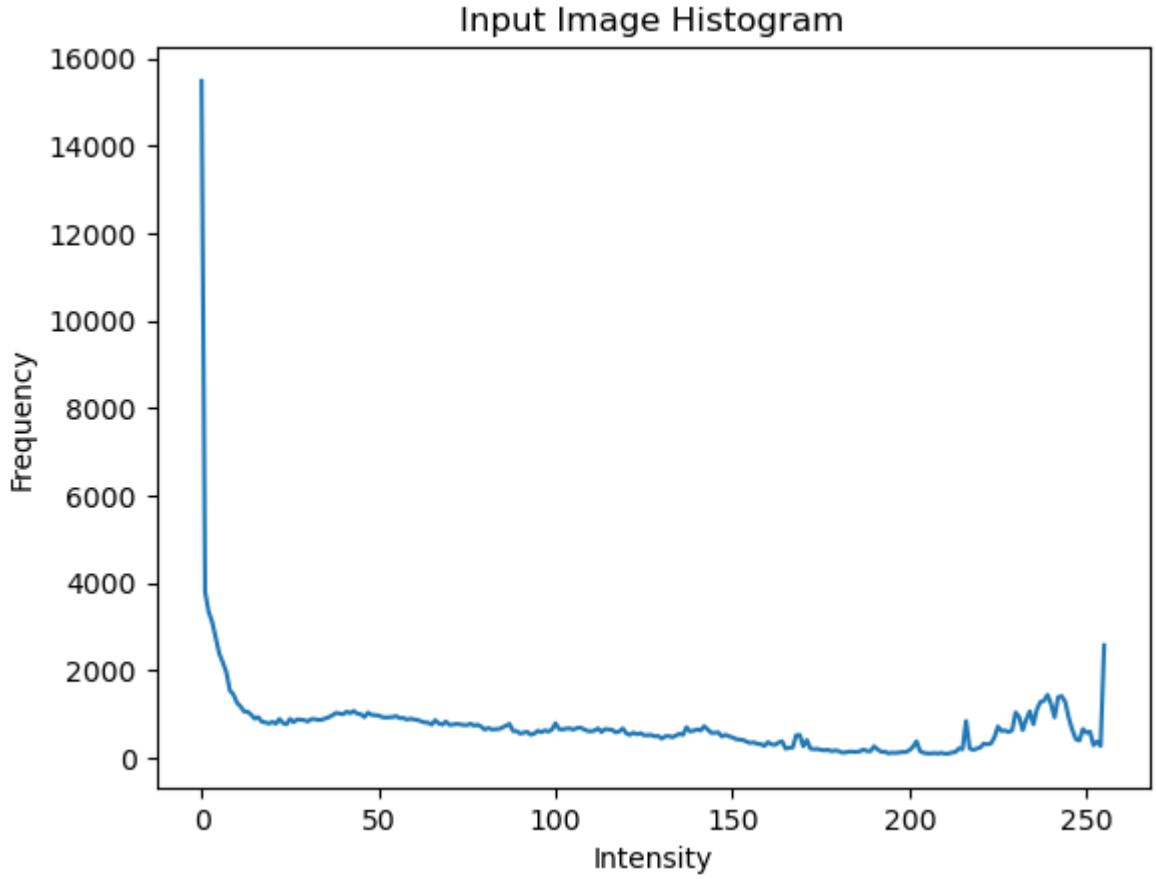
```
pip install opencv-python
import cv2
import numpy as np
import matplotlib.pyplot as plt
from skimage.exposure import match_histograms
def generate_histogram(image):
  histogram = cv2.calcHist([image], [0], None, [256], [0, 256])
  return histogram

def perform_histogram_equalization(image):
  gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
  # Apply histogram equalization
  equalized_image = cv2.equalizeHist(gray_image)
  return equalized_image

def perform_histogram_matching(image, reference_image):
  gray_input = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
  gray_reference = cv2.cvtColor(reference_image, cv2.COLOR_BGR2GRAY)
# Calculate histograms
  input_hist = generate_histogram(gray_input)
  reference_hist = generate_histogram(gray_reference)
# Perform histogram matching
  matched_image = match_histograms(gray_input, gray_reference)
  matched_image = np.uint8(matched_image)
  return matched_image

# Load the input image
image = cv2.imread('images/image_1.jpg')
# Generate and display the histogram of the input image
input_hist = generate_histogram(image)
plt.plot(input_hist)
plt.title("Input Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")
plt.show()
```

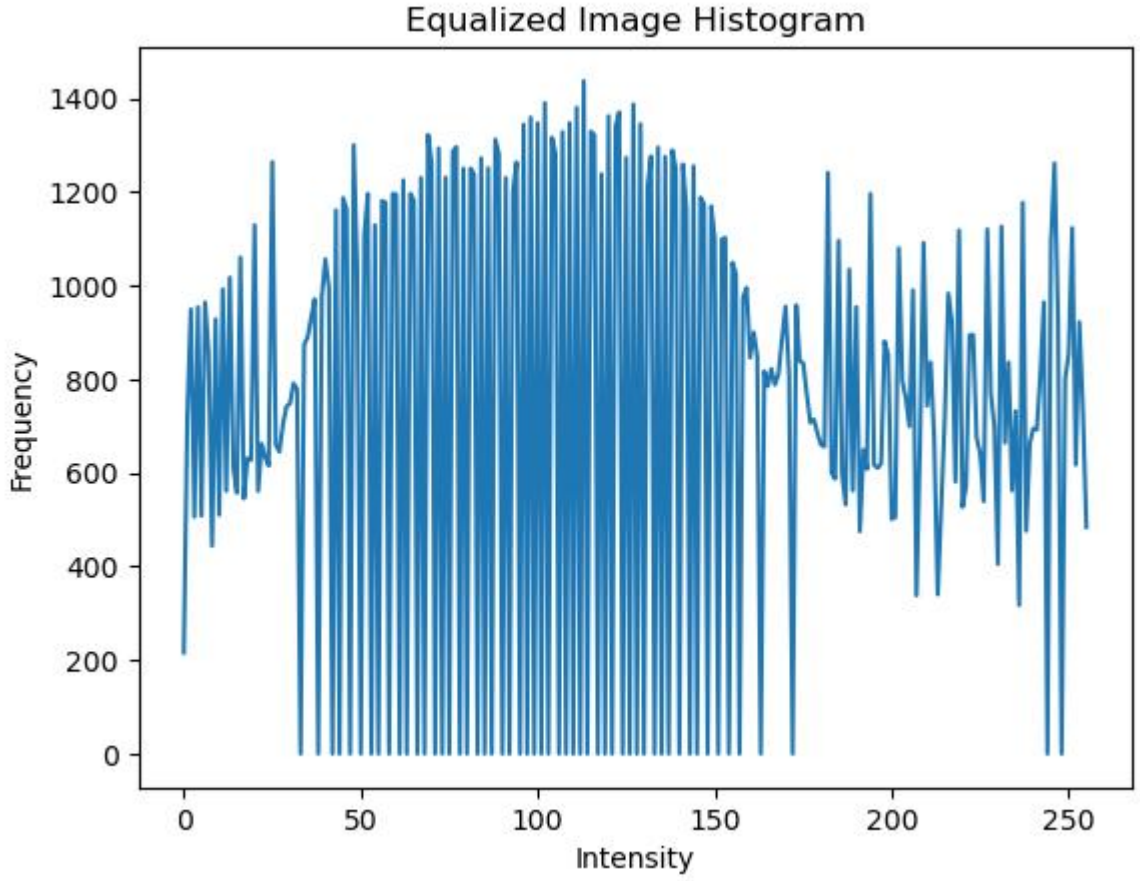
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Perform histogram equalization

```
equalized_image = perform_histogram_equalization(image)
# Generate and display the histogram of the equalized image
equalized_hist = generate_histogram(equalized_image)
plt.plot(equalized_hist)
plt.title("Equalized Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")
plt.show()
```

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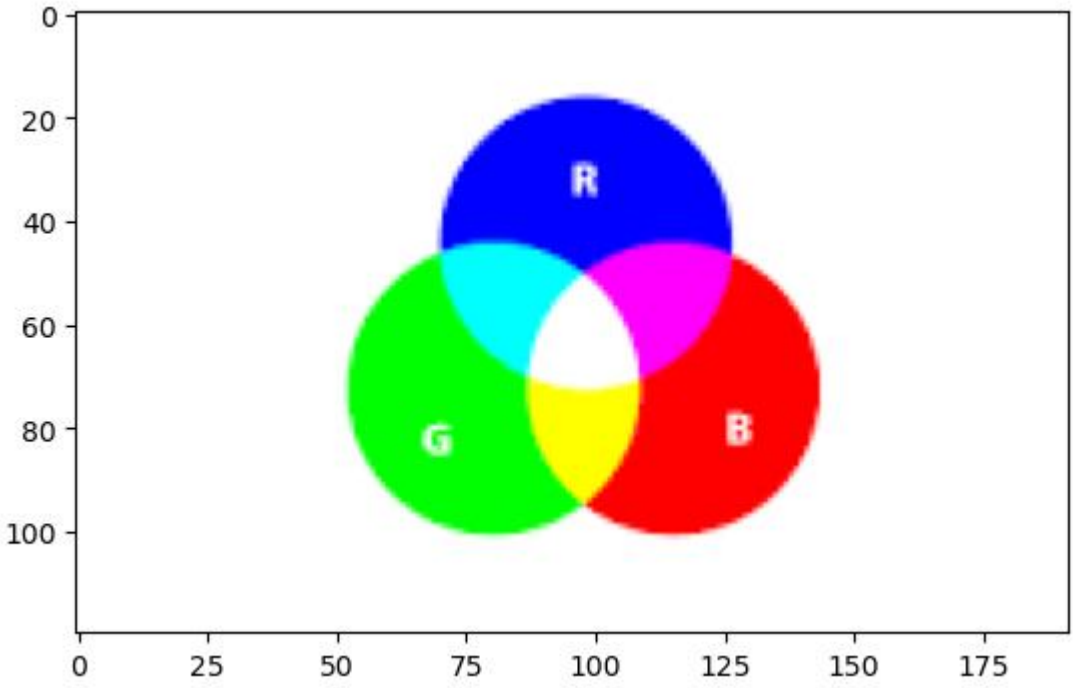


Load the reference image for histogram matching

reference_image = cv2.imread('images/images.png')

plt.imshow(reference_image)

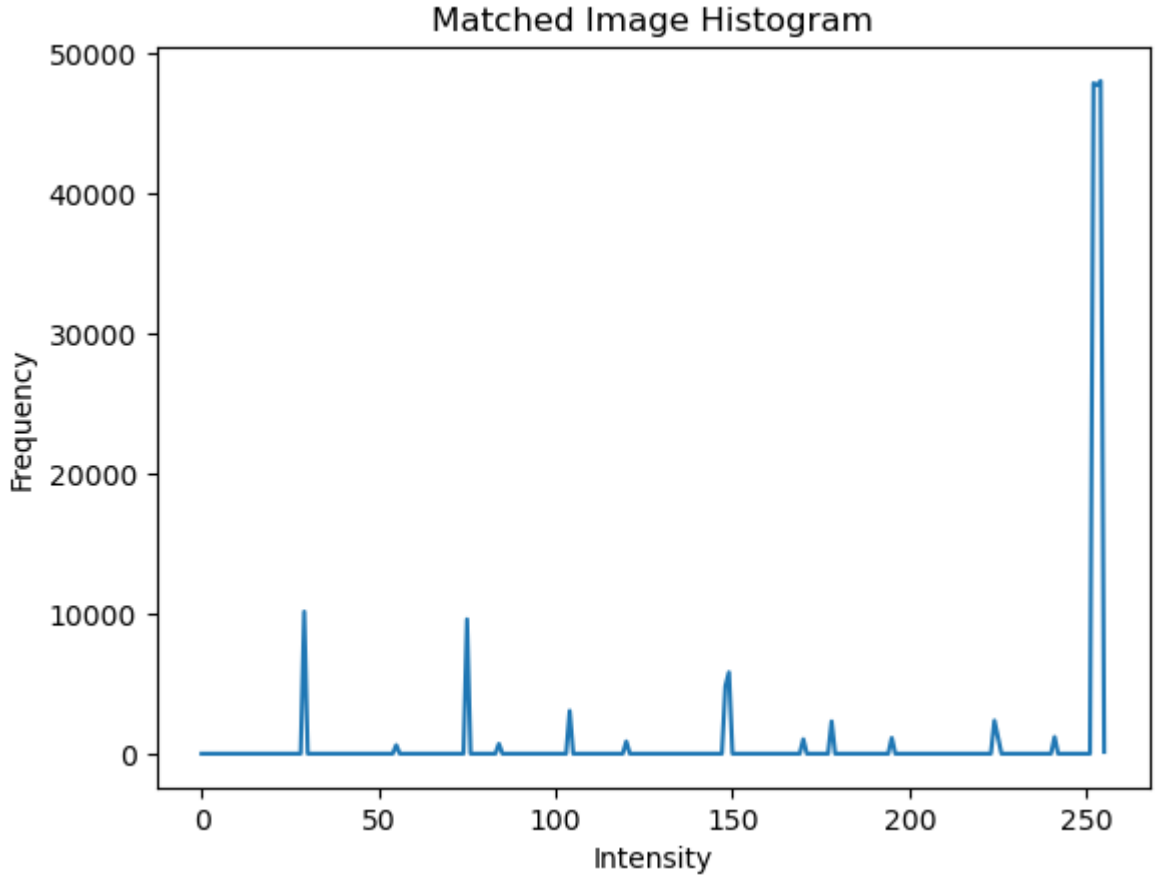
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```
# Perform histogram matching
matched_image = perform_histogram_matching(image, reference_image)
# Generate and display the histogram of the matched image
matched_hist = generate_histogram(matched_image)

plt.plot(matched_hist)
plt.title("Matched Image Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")
plt.show()
```

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```
plt.figure(figsize=(15,5))

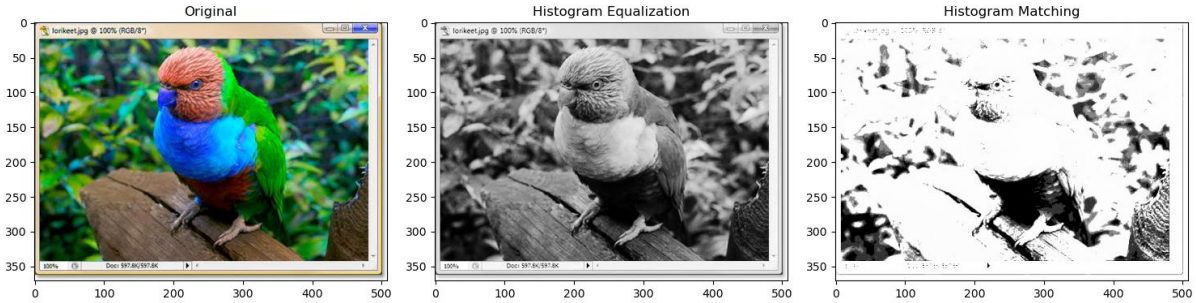
plt.subplot(1,3,1)
plt.imshow(image)
plt.title("Original")

plt.subplot(1,3,2)
plt.imshow(equalized_image,cmap='gray')
plt.title("Histogram Equalization")

plt.subplot(1,3,3)
plt.imshow(matched_image,cmap='gray')
plt.title("Histogram Matching")

plt.tight_layout()
plt.show()
```

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```
# Save the equalized and matched images
equalized_path = 'equalized.jpg'
matched_path = 'matched.jpg'
cv2.imwrite(equalized_path, equalized_image)
cv2.imwrite(matched_path, matched_image)
print(f"Equalized image saved at: {equalized_path}")
print(f"Matched image saved at: {matched_path}")
```

```
Equalized image saved at: equalized.jpg
Matched image saved at: matched.jpg
```

Conclusion

History Equalization

The original image first converted to grayscale image and then equalization is applied which stretched the dynamic range.
The global contrast is successfully balanced making textured much sharper.

History Matching

The resulting image has overexposure as most of the regions have been blown out as the reference image selected is highly polarized.